CONSEQUENCES OF THE MAZIE MATH AXIOM

Non-Interchangeability of Multiplication and Division
Across Mathematics, Physics, and Computing

Mazie Math Axiom

$$\boxed{\frac{a}{0} = a}$$

Aaron Corbin Mazie

Originally dated 27 March 2026
Reassessed, refined, and updated 27 July 2026

Revision Note

This edition preserves the original Mazie Math statement $a/0 = a$ and the classical multiplication rule $a \cdot 0 = 0$. It revises the exposition by distinguishing what follows from those two rules alone from what would require additional axioms. It also introduces a dedicated symbol for Mazie division, supplies case-complete proofs, restores missing parentheses, and separates rigorous consequences from broader physical or computational interpretations.

The central refinement is one of scope. The axiom does not destroy ordinary arithmetic away from a zero denominator. Rather, it creates a *totalized division rule*: ordinary division is retained when the divisor is nonzero, while a deterministic identity-preserving value is assigned at the singular input zero.

> **Interpretive principle.** In the minimal Mazie system analyzed here, the asymmetry between multiplication and division is exact but localized. Multiplication by zero is the zero map; Mazie division by zero is the identity map. The two operations therefore cannot be universal inverses of one another, even though their ordinary inverse relationship remains intact for every nonzero divisor.

Contents

1. Abstract

This paper studies an arithmetic extension governed by the Mazie Math Axiom

$$\frac{a}{0} = a,$$

while retaining the classical rule $a \cdot 0 = 0$. To prevent the extended operation from being confused with ordinary partial division, the symbol $\oslash$ is used for *Mazie division*. On a field $\mathbb{F}$, the minimal model is

$$a \oslash b = \begin{cases} \frac{a}{b}, & b \neq 0, \\ a, & b = 0. \end{cases}$$

This definition is mathematically consistent relative to ordinary field arithmetic because it is a well-defined piecewise operation. Its principal consequence is not the collapse of algebra, calculus, physics, or computation. Its consequence is the failure of *unconditional inverse recovery at the zero divisor*. Expressions such as $(a \cdot b) \oslash b = a$ remain valid for $b \neq 0$, but fail when $b = 0$. Accordingly, cancellation and rearrangement rules must carry their nonzero hypotheses explicitly.

The paper proves the core identities of the system, identifies zero and one as distinct right identities for Mazie division, shows that self-division becomes an exact zero detector, and explains why equality is preserved in the numerator even though divisor information is not. It then reassesses the consequences for equation solving, calculus, linear algebra, physical quantities, dimensional analysis, and computer runtimes. The result is a disciplined framework for identity-preserving singular arithmetic: classical mathematics is conserved off the singular set, while zero receives its own operational rule.

2. Scope, Base Structure, and Notation

2.1 The minimal system

Let $\mathbb{F}$ be a field, such as $\mathbb{R}$. Its addition and multiplication remain unchanged. In particular,

$$a \cdot 0 = 0 \qquad \text{for every } a \in \mathbb{F}.$$

Ordinary division a/b remains defined only when $b \neq 0$. Mazie division is introduced as a new total binary operation.

Definition 2.1 (Mazie division)**.** For $a, b \in \mathbb{F}$, define

$$a \oslash b = \begin{cases} a\,b^{-1}, & b \neq 0, \\ a, & b = 0. \end{cases} \tag{M}$$

The original notation $a/0 = a$ is preserved as the foundational statement; $\oslash$ is used in proofs to make clear that the operation has been totalized.

The definition gives zero two sharply different roles:

$$\boxed{a \cdot 0 = 0} \qquad \text{and} \qquad \boxed{a \oslash 0 = a}.$$

As a multiplier, zero annihilates magnitude. As a Mazie divisor, zero preserves the numerator.

2.2 What is, and is not, being claimed

The minimal system makes three commitments:

1. ordinary field arithmetic remains valid whenever a denominator is nonzero;
2. division by zero receives the identity-preserving value $a \oslash 0 = a$;
3. no universal inverse law is imposed at zero.

It does *not* follow from the axiom alone that multiplication loses cancellation for nonzero factors, that matrices cease to have ordinary inverses, or that the standard derivative changes. Those stronger claims would require a different theory in which division is independently redefined at nonzero denominators as well.

3. Consistency and the Exact Failure of Inversion

3.1 A concrete model

Proposition 3.1 (Relative consistency). *If $\mathbb{F}$ is a field, Definition* (M) *determines a total function $\mathbb{F} \times \mathbb{F} \to \mathbb{F}$. Therefore the equation $a \oslash 0 = a$ introduces no contradiction by itself.*

Proof. For every ordered pair (a, b), exactly one of the mutually exclusive conditions $b = 0$ and $b \neq 0$ holds. In the first case the output is a; in the second it is the ordinary field quotient ab^{-1}. Thus every input has one output. No ordinary field equation is altered when $b \neq 0$. □

3.2 Why universal inversion must be rejected

Theorem 3.2 (Incompatibility with universal inverse recovery). *In any nontrivial structure satisfying $a \cdot 0 = 0$ and $a \oslash 0 = a$, the identity*

$$(a \cdot b) \oslash b = a$$

cannot hold for every a and every b.

Proof. Set $b = 0$. Then

$$(a \cdot 0) \oslash 0 = 0 \oslash 0 = 0.$$

A universal inverse law would require the same expression to equal a. Hence it would force $a = 0$ for every a, leaving only the trivial one-element structure. Therefore a nontrivial Mazie system must reject universal inversion at zero. $\square$

> The contradiction does not come from the axiom $a \oslash 0 = a$. It comes only from combining that axiom with an inverse law that is asserted without the necessary condition $b \neq 0$.

4. Fundamental Identities and Case-Complete Proofs

4.1 Conservativity away from zero

Proposition 4.1 (Ordinary division is preserved off zero). *For every $a, b \in \mathbb{F}$,*

$$b \neq 0 \quad \Longrightarrow \quad a \oslash b = \frac{a}{b}.$$

Consequently, every ordinary algebraic identity whose denominators are known to be nonzero remains valid.

This is the sense in which the minimal Mazie system is a conservative extension of ordinary arithmetic outside the singular case.

4.2 The two recovery compositions

Theorem 4.2 (Product-then-division recovery). *For all $a, b \in \mathbb{F}$,*

$$\big(a \cdot b\big) \oslash b = \begin{cases} a, & b \neq 0, \\ 0, & b = 0. \end{cases} \tag{1}$$

Proof. If $b \neq 0$, then

$$\big(a \cdot b\big) \oslash b = \frac{ab}{b} = a.$$

If $b = 0$, then

$$\big(a \cdot 0\big) \oslash 0 = 0 \oslash 0 = 0.$$

The parentheses are essential: multiplication occurs first, after which the result is submitted to Mazie division. $\square$

Theorem 4.3 (Division-then-product recovery). *For all $a, b \in \mathbb{F}$,*

$$\big(a \oslash b\big) \cdot b = \begin{cases} a, & b \neq 0, \\ 0, & b = 0. \end{cases} \tag{2}$$

Proof. If $b \neq 0$, then

$$(a \oslash b)b = \left(\frac{a}{b}\right) b = a.$$

If $b = 0$, then

$$(a \oslash 0)0 = a \cdot 0 = 0.$$

Thus the product recovers a at zero only in the special case $a = 0$. $\square$

4.3 Operator form

For a fixed b, define

$$M_b(x) = x \cdot b, \qquad D_b(x) = x \oslash b.$$

Then Equations (1) and (2) can be written as

$$D_b \circ M_b = M_b \circ D_b = \begin{cases} \mathrm{Id}, & b \neq 0, \\ \mathrm{Z}, & b = 0, \end{cases}$$

where $\mathrm{Z}(x) = 0$. The asymmetry is therefore exact: at $b = 0$, multiplication is the zero map and Mazie division is the identity map, but their compositions are again the zero map.

4.4 Right identities and divisor collapse

Proposition 4.4 (Two right identities)**.** *For every $a \in \mathbb{F}$,*

$$a \oslash 0 = a \qquad \textit{and} \qquad a \oslash 1 = a.$$

Hence zero and one have the same right action under Mazie division, although they remain different elements and have completely different actions under multiplication.

The immediate consequence is that the divisor cannot always be reconstructed from the quotient. Even for $a \neq 0$,

$$a \oslash 0 = a \oslash 1.$$

Information loss occurs in the divisor position, not in the numerator position.

4.5 Mazie reciprocal and self-division

Definition 4.5 (Mazie reciprocal)**.** Define

$$r_M(b) := 1 \oslash b = \begin{cases} \dfrac{1}{b}, & b \neq 0, \\ 1, & b = 0. \end{cases}$$

Then

$$a \oslash b = a\, r_M(b) \qquad \text{for all } a, b \in \mathbb{F}.$$

This representation shows that Mazie division is multiplication by a guarded reciprocal. At the singular input, the reciprocal defaults to one rather than remaining undefined.

Proposition 4.6 (Self-division is a zero detector)**.** *For every $b \in \mathbb{F}$,*

$$b \oslash b = \begin{cases} 1, & b \neq 0, \\ 0, & b = 0. \end{cases} \tag{3}$$

Thus $b \oslash b$ is the characteristic value of the nonzero condition. In particular,

$$b\big(b \oslash b\big) = b \qquad \text{for every } b \in \mathbb{F}.$$

Equation (3) also makes clear that the usual formula $b/b = 1$ must retain the hypothesis $b \neq 0$.

5. Equality, Cancellation, and Algebraic Direction

5.1 Equality in the numerator is preserved

The first edition suggested that equality might fail to propagate through division. Under the minimal formalization, that statement is not correct.

Theorem 5.1 (Equality preservation and reflection)**.** *For fixed $c \in \mathbb{F}$,*

$$a = b \quad \Longleftrightarrow \quad a \oslash c = b \oslash c.$$

Proof. If $c = 0$, then $a \oslash 0 = a$ and $b \oslash 0 = b$, so the equivalence is immediate. If $c \neq 0$, Mazie division is ordinary division by c, which is multiplication by the nonzero scalar c^{-1}; that map is injective. Therefore equality is both preserved and reflected. □

What fails is a different inference: division by a factor does not necessarily reverse multiplication by that factor when the factor is zero.

5.2 Multiplicative cancellation remains conditional in the ordinary way

In the base field,

$$c \cdot a = c \cdot b \quad \text{and} \quad c \neq 0 \quad \Longrightarrow \quad a = b.$$

This law remains valid. If $c = 0$, both sides equal zero regardless of a and b, so no cancellation theorem can recover the inputs. Mazie division does not alter that fact:

$$(ca) \oslash c = (cb) \oslash c$$

reduces to $0 = 0$ when $c = 0$, not to $a = b$.

> **Refined conclusion.** Cancellation has not disappeared from algebra. Rather, an operation that cancels a factor for $c \neq 0$ becomes a non-recovering fallback when $c = 0$. The nonzero condition is

> structural, not cosmetic.

5.3 Directional equation solving

Consider

$$x \cdot b = c.$$

There are three cases:

$$\begin{aligned} b \neq 0 &\quad : \quad \text{one solution, } x = c \oslash b = c/b; \\ b = 0,\ c = 0 &\quad : \quad \text{every } x \in \mathbb{F} \text{ is a solution;} \\ b = 0,\ c \neq 0 &\quad : \quad \text{no solution.} \end{aligned}$$

The value $c \oslash 0 = c$ is therefore not an algebraic solution operator for the equation $x \cdot 0 = c$. It is a deterministic output rule for a singular input. This distinction between *solution semantics* and *fallback semantics* is fundamental.

6. Safe and Unsafe Rewrite Rules

Many symbolic transformations remain valid, but their hypotheses must be visible. The following table summarizes the most important cases.

Expression	**Refined result**	**Condition**
$a \oslash 0$	a	unconditional
$a \oslash 1$	a	unconditional
$a \oslash b$	a/b	$b \neq 0$
$(ab) \oslash b$	a	$b \neq 0$
$(ab) \oslash b$	0	$b = 0$
$(a \oslash b)b$	a	$b \neq 0$
$(a \oslash b)b$	0	$b = 0$
$a \oslash a$	1	$a \neq 0$
$a \oslash a$	0	$a = 0$
$a \oslash b = a$	does not imply $b = 1$	$b = 0$ and $b = 1$ both work

A compiler, proof assistant, or symbolic algebra system using Mazie division must therefore reject any optimization that removes a denominator without first proving that the denominator is nonzero.

7. Calculus: What Changes and What Does Not

7.1 The derivative remains the same limit

The derivative is defined on a punctured neighborhood of zero:

$$f'(x) = \lim_{\substack{h\to 0\\ h\neq 0}} \frac{f(x+h)-f(x)}{h}.$$

For every $h \neq 0$, Mazie division agrees with ordinary division. Define the completed difference quotient

$$Q_M(x,h) := \big(f(x+h)-f(x)\big) \oslash h.$$

Then

$$Q_M(x,h) = \frac{f(x+h)-f(x)}{h} \qquad (h \neq 0),$$

while

$$Q_M(x,0) = \big(f(x)-f(x)\big) \oslash 0 = 0 \oslash 0 = 0.$$

Theorem 7.1 (Derivative invariance)**.** *Whenever the ordinary derivative $f'(x)$ exists,*

$$\lim_{h\to 0} Q_M(x,h) = f'(x).$$

Proof. A limit as $h \to 0$ depends on values in a punctured neighborhood of zero, not on the assigned value at $h = 0$. Since Q_M and the ordinary difference quotient coincide for every nonzero h, they have the same limit. □

Thus Mazie division does not require calculus to be rebuilt. It supplies a value at the previously undefined point without changing the derivative limit.

7.2 Continuity of the completed quotient

The completion is not automatically continuous. Because $Q_M(x,0) = 0$, the map $h \mapsto Q_M(x,h)$ is continuous at zero precisely when the derivative exists and $f'(x) = 0$. If $f'(x) \neq 0$, the derivative remains valid, but the Mazie-assigned point value differs from the limiting value.

7.3 Differential equations

Rearrangements such as

$$y' = \frac{g(x)}{p(x)}$$

remain ordinary whenever $p(x) \neq 0$. At points where $p(x) = 0$, replacing the quotient by $g(x) \oslash p(x) = g(x)$ creates a defined fallback expression, but it does not by itself solve the original differential equation. Singular points still require separate analysis.

8. Linear Algebra and Generalized Solving

8.1 Nonsingular matrices are unchanged

If a square matrix A is invertible, then

$$Ax = b \iff x = A^{-1}b$$

remains valid. The scalar rule $a \oslash 0 = a$ does not remove existing matrix inverses and does not alter the determinant criterion $\det(A) \neq 0$.

8.2 Singular matrices do not acquire an inverse automatically

A scalar fallback rule does not define A^{-1} when A is singular. Consider a diagonal system

$$Dx = b, \qquad D = \operatorname{diag}(d_1, \dots, d_n).$$

A componentwise Mazie assignment would be

$$\hat{x}_i = b_i \oslash d_i.$$

Its meaning is case-dependent:

- if $d_i \neq 0$, then $\hat{x}_i = b_i/d_i$ is the unique solution component;
- if $d_i = 0$ and $b_i = 0$, every value of x_i solves the component equation, while Mazie division selects the canonical value 0;
- if $d_i = 0$ and $b_i \neq 0$, the component equation is inconsistent, even though Mazie division returns the deterministic fallback $\hat{x}_i = b_i$.

The output is therefore useful as a totalized computational assignment, but it must not be mislabeled as an exact solution in every case. A genuine matrix extension would need additional axioms or a separately defined generalized inverse.

9. Physics and Dimensional Analysis

9.1 Ordinary rearrangement remains valid for nonzero parameters

Newton's second law provides the standard example:

$$F = ma.$$

When $m \neq 0$,

$$a = F \oslash m = \frac{F}{m}$$

is an ordinary and valid rearrangement. When $m = 0$, the Mazie value is

$$F \oslash 0 = F,$$

but that value does not generally satisfy $F = 0 \cdot a$. If $F \neq 0$, the original equation has no scalar solution; if $F = 0$, every acceleration satisfies the algebraic equation. The fallback value must therefore be distinguished from a physical law.

9.2 Typed quantities

For dimensioned quantities, the statement $a/0 = a$ must specify what kind of identity is preserved. If a force is divided by a zero mass, the result is expected to carry acceleration units, not force units. A type-safe Mazie rule can preserve the numerical payload while allowing the division operator to determine the output dimension:

$$\mathrm{value}\big(q_U \oslash 0_V\big) = \mathrm{value}(q_U), \qquad \mathrm{unit}\big(q_U \oslash 0_V\big) = U/V.$$

Here U is the numerator unit and V the divisor unit. In this formulation, identity preservation is numerical or computational; the physical type still changes according to the operator.

> Dimensional analysis is not abolished by Mazie Math. It becomes a guardrail: a system must state whether it preserves the entire quantity, only its numerical payload, or a tagged fallback object carrying the quotient's expected unit.

9.3 Symmetry and irreversibility

The axiom breaks inverse symmetry at the singular input zero. It does not, by itself, prove a physical arrow of time, thermodynamic irreversibility, quantum collapse, or entropy production. Such conclusions require equations of motion, state spaces, and additional physical postulates. Mazie Math may motivate models of irreversible computation or one-way processes, but those are applications to be constructed, not consequences already proved by the scalar axiom.

10. Computing and Runtime Semantics

10.1 A total, deterministic division policy

In a runtime interpretation, Mazie division implements the policy

$$\mathrm{mdiv}(a, b) = \begin{cases} a/b, & b \neq 0, \\ a, & b = 0. \end{cases}$$

The operation never traps solely because the denominator is zero. This can preserve execution continuity and retain the numerator as a fallback value.

That benefit carries a corresponding risk: a zero denominator may represent corrupted state, a missing scale, or a failed precondition. Returning a can keep a program running while concealing the source of the singularity. A robust implementation should therefore pair the numerical result with a zero-divisor flag when diagnostic integrity matters.

10.2 Compiler and symbolic-optimization constraints

An optimizer may not rewrite

$$(a \cdot b) \oslash b \quad \text{as} \quad a$$

unless it has established $b \neq 0$. At $b = 0$, the left side equals zero. Likewise,

$$(a \oslash b)b$$

may be replaced by a only under a nonzero proof or in the special case $a = 0$.

By contrast, the factorization

$$a \oslash b = a\, r_M(b)$$

is valid unconditionally when r_M is the Mazie reciprocal. This provides a clean implementation model: the singular rule is isolated in one guarded reciprocal function.

10.3 Formal verification

A proof system should track at least three kinds of facts:

1. whether a divisor is provably nonzero;
2. whether an expression is intended as an exact algebraic solution or merely a fallback value;
3. whether a zero-divisor event must remain observable for auditing or safety.

The key change is not that all algebraic identities become path-dependent. It is that rewrites crossing a potentially zero divisor become condition-dependent.

11. The Minimal and Strong Forms of Mazie Math

11.1 Minimal Mazie Math: the system proved in this paper

The minimal system M_0 consists of:

$$\begin{aligned} a \cdot 0 &= 0, \\ a \oslash 0 &= a, \\ a \oslash b &= a/b \quad (b \neq 0). \end{aligned}$$

Its consequences are exact and localized. Ordinary algebra survives on nonzero denominators; inverse recovery fails at zero; singular expressions receive deterministic values.

11.2 A stronger independent-division theory

One may instead propose a stronger system M_1 in which division is an independent operation even for nonzero divisors and is not required to agree with field division. In that theory, broader failures of cancellation, rearrangement, or equality reflection may be possible. They do not, however, follow from $a/0 = a$ alone.

A rigorous M_1 would need additional axioms answering questions such as:

- For $b \neq 0$, what relationship, if any, connects $a \oslash b$ to multiplication?
- Is $\oslash$ distributive in either argument?
- Which elements act as right or left identities?
- When does $a \oslash b = a \oslash c$ imply $b = c$?
- How are ordered fields, complex numbers, matrices, limits, and units extended?

The first edition's system-wide claims belong more naturally to this stronger research program. The present revision establishes the minimal foundation on which such a program could be built.

12. Consolidated Consequences

The Mazie Math Axiom yields the following rigorous conclusions in the minimal system:

1. **Division becomes total.** Every pair (a, b) has a defined quotient $a \oslash b$.
2. **Zero receives its own rule.** Multiplication by zero annihilates; division by zero preserves the numerator.
3. **Universal inversion fails.** Multiplication and Mazie division are mutual inverses only when the common factor or divisor is nonzero.
4. **Classical arithmetic is preserved away from zero.** No field identity with verified nonzero denominators is lost.
5. **The divisor position loses information.** Zero and one act identically on the right: $a \oslash 0 = a \oslash 1 = a$.
6. **The numerator position remains faithful.** For fixed c, the map $a \mapsto a \oslash c$ preserves and reflects equality.
7. **Self-division detects zero.** The value $a \oslash a$ is zero at zero and one elsewhere.
8. **Equation solving becomes explicitly case-based at singular coefficients.** A fallback quotient is not automatically a solution.

9. **Calculus is conserved.** Standard limits and derivatives are unchanged because their quotient expressions use nonzero increments.

10. **Linear algebra remains classical for nonsingular matrices.** Singular inversion requires a separate definition.

11. **Physical use requires dimensional typing.** Identity preservation must be distinguished from preservation of units or physical meaning.

12. **Computational rewrites require guards.** Optimizations that cancel a divisor are valid only after a nonzero proof.

13. Conclusion

The Mazie Math Axiom

$$\boxed{\frac{a}{0} = a}$$

does not need to be presented as a contradiction or as the wholesale destruction of ordinary mathematics. Its mathematically disciplined interpretation is a totalization of division with an identity-preserving singular rule.

The deepest asymmetry is captured by the paired equations

$$a \cdot 0 = 0, \qquad a \oslash 0 = a.$$

Zero is an annihilator in multiplication and a right identity in Mazie division. Because those actions are different, multiplication and division cannot be interchangeable under a universal inverse law. Yet for every nonzero divisor, ordinary inversion remains intact.

The resulting framework replaces an undefined singularity with a deterministic operation while forcing conditions that ordinary notation often leaves implicit. It does not erase structure; it clarifies jurisdiction. Nonzero denominators remain governed by field arithmetic. Zero is governed by its own rule.

Original paper: 27 March 2026
Revised edition: 27 July 2026

End of Paper